UChicago REU Notes Week 5

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Contents

To be honest, last week's note isn't as successful, so we'll mainly focus on the category theory from now on. Here, we assume that G is a finite group, and the categories (mentioned as objects of Cat or similar objects) are small.

1 Statements on G-categories

This part is devoted to understand some basics. I'll fill in the details for preliminaries on G-categories later, together with adjunction $(\mathrm{Ob} \dashv \mathcal{E})$ between Cat and Set.

Given a category $\mathcal{A}$, the functor category $\mathsf{Cat}(\mathcal{E}(\mathcal{G}), \mathcal{A})$ is automatically a G-category, as each element $g \in G$ can be realized as an automorphism functor on the indiscrete category $\mathcal{E}G$, so it admits both a left/right action by precomposing either g^{-1} or g as automorphism of $\mathcal{E}G$.

Similarly, if $\mathcal{A}$ itself already has a left/right G-action, then it defines a left/right G-action on $\mathsf{Cat}(\mathcal{B}, \mathcal{A})$ by postcomposition; similarly if $\mathcal{B}$ is a G-category, then precomposition also defines one. Combining the two, it admits a conjugation action on $\mathsf{Cat}(\mathcal{B}, \mathcal{A})$ if both $\mathcal{B}, \mathcal{A}$ admits a left action.

Given G-categories $\mathcal{A}, \mathcal{B}$, denote $\mathsf{Cat}_G(\mathcal{B}, \mathcal{A}) := \mathsf{Cat}(\mathcal{B}, \mathcal{A})$, and $G\mathsf{Cat}(\mathcal{B}, \mathcal{A}) := \mathsf{Cat}(\mathcal{B}, \mathcal{A})^G$ as the G-fixpoint (i.e. all functors commuting with G-actions).

Proposition 1.1. *There is a G-equivarant isomorphism when giving $\mathcal{A}, \mathcal{B}, \mathcal{C}$ three G-categories:*

$$\Phi : \mathsf{Cat}_G(\mathcal{A} \times \mathcal{B}, \mathcal{C}) \cong \mathsf{Cat}_G(\mathcal{A}, \mathsf{Cat}(\mathcal{B}, \mathcal{C}))$$

Proof. We recognize this as a non-equivariant statement first, then show if all three are G-categories (with G acts diagonally on $\mathcal{A} \times \mathcal{B}$), such isomorphism is G-equivariant.

I. Construction of Functor:

Given a functor $F : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$, for any object $X \in \mathcal{A}$ (together with identity $\mathrm{id}_X : X \to X$), define a functor $F_X : \mathcal{B} \to \mathcal{C}$ by $F_X(Y) := F(X, Y)$, and given any morphism $g : Y \to Y'$ in $\mathcal{B}$, define the morphism $F_X g : F_X(Y) \to F_X(Y')$ as the morphism $F(\mathrm{id}_X, g) : F(X, Y) \to F(X, Y')$. The routine check that this is functorial, using the product category definition:

- Given $\mathrm{id}_Y : Y \to Y$, we have $F_X \mathrm{id}_Y = F(\mathrm{id}_X, \mathrm{id}_Y) = \mathrm{id}_{F(X,Y)} = \mathrm{id}_{F_X(Y)}$.

- Given $g : Y \to Y'$, $g' : Y' \to Y''$, we have

$$F_X(g' \circ g) = F(\mathrm{id}_X, g' \circ g) = F(\mathrm{id}_X, g') \circ F(\mathrm{id}_X, g) = F_X g' \circ F_X g$$

Now, given a morphism $f : X \to X'$ in $\mathcal{A}$, we shall define a natural transformation $Nf : F_X \to F_{X'}$. Realistically, for any $Y \in \mathcal{B}$, the only choice is by $Nf_Y := F(f, \mathrm{id}_Y) : F(X,Y) \to F(X',Y)$. Which, given $g : Y \to Y'$ in $\mathcal{B}$, the following diagram commutes using product category property:

$$\begin{tikzcd}
F(X,Y) \arrow[r, "F(\mathrm{id}_X, g)"] \arrow[d, "F(f,\mathrm{id}_Y)"'] & F(X,Y') \arrow[d, "F(f,\mathrm{id}_{Y'})"] & \equiv & F_X(Y) \arrow[r, "F_X g"] \arrow[d, "Nf_Y"'] & F_X(Y') \arrow[d, "Nf_{Y'}"] \\
F(X',Y) \arrow[r, "F(\mathrm{id}_{X'}, g)"'] & F(X',Y') & & F_{X'}(Y) \arrow[r, "F_{X'} g"'] & F_{X'}(Y')
\end{tikzcd}$$

Hence, $Nf : F_X \to F_{X'}$ is a natural transformation.

Then, we claim that $N : \mathcal{A} \to \mathsf{Cat}(\mathcal{B}, \mathcal{C})$ by $N(X) := F_X$, $Nf : F_X \to F_{X'}$ as above is a functor:

- $(N\mathrm{id}_X)_Y = F(\mathrm{id}_X, \mathrm{id}_Y) = \mathrm{id}_{F(X,Y)} = \mathrm{id}_{F_X(Y)}$, so $N\mathrm{id}_X = \mathrm{id}_{F_X}$.
- Given $f : X \to X'$, $f' : X' \to X''$, then

$$N(f' \circ f)_Y = F(f' \circ f, \mathrm{id}_Y) = F(f', \mathrm{id}_Y) \circ F(f, \mathrm{id}_Y) = Nf'_Y \circ Nf_Y$$

So, $N(f' \circ f) = Nf' \circ Nf$.

Now, suppose two such functors $F, F' : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$ has a natural transformation $\mu : F \to F'$, we want to construct a natural transformation $\overline{\mu} : N \to N'$, where $N, N' : \mathcal{A} \to \mathsf{Cat}(\mathcal{B}, \mathcal{C})$ are functors associated to F, F'. Which, given any $X \in \mathcal{A}$, we see the collection $(\overline{\mu}_X)_Y := \mu_{(X,Y)}$ is a natural transformation $F_X \to F'_X$, due to the following diagram for all $g : Y \to Y'$ in $\mathcal{B}$:

$$\begin{tikzcd}
F(X,Y) \arrow[r, "F(\mathrm{id}_X, g)"] \arrow[d, "\mu_{(X,Y)}"'] & F(X,Y') \arrow[d, "\mu_{(X,Y')}"] & \equiv & F_X(Y) \arrow[r, "F_X g"] \arrow[d, "(\overline{\mu}_X)_Y"'] & F(X,Y') \arrow[d, "(\overline{\mu}_X)_{Y'}"] \\
F'(X,Y) \arrow[r, "F'(\mathrm{id}_X, g)"'] & F'(X,Y') & & F'_X(Y) \arrow[r, "F'_X g"'] & F'_X(Y')
\end{tikzcd}$$

This shows $\overline{\mu}_X : F_X \to F'_X$ is a natural transformation, or morphism within $\mathsf{Cat}(\mathcal{B}, \mathcal{C})$. Then, notice that if $\mu : F \to F'$, $\mu' : F' \to F''$ are two natural transformations between functors $F, F', F'' : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$, the following holds:

$$(\overline{\mu' \circ \mu}_X)_Y = (\mu' \circ \mu)_{(X,Y)} = \mu'_{(X,Y)} \circ \mu_{(X,Y)} = (\overline{\mu'}_X)_Y \circ (\overline{\mu}_X)_Y$$

This shows as natural transformations, $\overline{\mu}_X : N_X \to N'_X$ and $\overline{\mu'}_X : N'_X \to N''_X$ satisfies $(\overline{\mu' \circ \mu})_X = \overline{\mu'}_X \circ \overline{\mu}_X$, so $\overline{\mu' \circ \mu} = \overline{\mu'} \circ \overline{\mu}$ as natural transformations $\overline{\mu} : N \to N'$, $\overline{\mu'} : N' \to N''$.

So, define $\Phi(F : \mathcal{A} \times \mathcal{B} \to \mathcal{C}) := N : \mathcal{A} \to \mathsf{Cat}(\mathcal{B}, \mathcal{C})$, and $\Phi(\mu : F \to F') := \overline{\mu} : N \to N'$ as above, it's a legitimate functor.

II. Isomorphism on Objects:

Suppose $\Phi(F) = \Phi(F')$, then it's clear F, F' have the same action on the objects (as all $X \in \mathcal{A}$ satisfies $F_X = F'_X$, so all $Y \in \mathcal{B}$ satisfies $F(X,Y) = F_X(Y) = F'_X(Y) = F'(X,Y)$). Now, given $f : X \to X'$ in $\mathcal{A}$ and $g : Y \to Y'$ in B, we also see that:

$$F(f,g) = F(f, \mathrm{id}_{Y'}) \circ F(\mathrm{id}_X, g) = Nf_{Y'} \circ F_X g = F' f_{Y'} \circ F'_X g = F'(f, \mathrm{id}_{Y'}) \circ F'(\mathrm{id}_X, g) = F'(f,g)$$

($Nf = N'f$ is by $\Phi(F) = \Phi(F')$, then $F_X = F'_X$ also the same logic).

This shows $F = F'$, as its map on morphisms are also the same.

Conversely, given a functor $G : \mathcal{A} \to \mathsf{Cat}(\mathcal{B}, \mathcal{C})$, we'd like to construct a functor $\tilde{G} : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$. Object wise, define $G(X, Y) := [G(X)](Y)$ (here, $G(X) : \mathcal{B} \to \mathcal{C}$ is a functor), and given morphisms $f : X \to X'$ in $\mathcal{A}$, $g : Y \to Y'$ in $\mathcal{B}$, define morphism $G(f, g) : G(X, Y) \to G(X', Y')$ as the following morphism in the commutative diagram:

$$\begin{tikzcd}
G(X,Y) := [G(X)](Y) \arrow[r, "G(X)g"] \arrow[d, "Gf_Y"'] & {[G(X)](Y')} \arrow[d, "Gf_{Y'}"] \\
{[G(X')](Y)} \arrow[r, "G(X')g"'] & {[G(X')](Y') =: G(X', Y')}
\end{tikzcd}$$

Here, recall that for each $f : X \to X'$, $Gf : G(X) \to G(X')$ is a morphism in $\mathsf{Cat}(\mathcal{B}, \mathcal{C})$ (i.e. a natural transformation), so the above diagram holds.

Now, to verify functoriality, consider the following:

- Given id_X in $\mathcal{A}$, id_Y in $\mathcal{B}$, we get $G(\mathrm{id}_X) = \mathrm{id}_{G(X)}$, so each natural transformation $G(\mathrm{id}_X)_Y = \mathrm{id}_{[G(X)](Y)}$. Similarly, since $G(X)$ is a functor, we get $G(X)\mathrm{id}_Y = \mathrm{id}_{[G(X)](Y)}$. So, the diagram becomes:

$$\begin{tikzcd}
\tilde{G}(X,Y) := [G(X)](Y) \arrow[r, "\mathrm{id}_{[G(X)](Y)}"] \arrow[d, "\mathrm{id}_{[G(X)](Y)}"'] & {[G(X)](Y)} \arrow[d, "\mathrm{id}_{[G(X)](Y)}"] \\
{[G(X)](Y)} \arrow[r, "\mathrm{id}_{[G(X)](Y)}"'] & {[G(X)](Y) =: \tilde{G}(X, Y)}
\end{tikzcd}$$

  Hence, $G(\mathrm{id}_X, \mathrm{id}_Y)$ results in the identity of $G(X, Y)$.

- Given $f : X \to X', f' : X' \to X''$ in $\mathcal{A}$, and $g : Y \to Y', g' : Y' \to Y''$ in $\mathcal{B}$, the following gigantic diagram holds, due to the functoriality $G(f' \circ f) = Gf' \circ Gf$, and each $G(X)(g' \circ g) = G(X)g' \circ G(X)g$:

$$\begin{tikzcd}
\tilde{G}(X,Y) := [G(X)](Y) \arrow[r, "G(X)g"] \arrow[d, "Gf_Y"'] & {[G(X)](Y')} \arrow[r, "G(X)g'"] \arrow[d, "Gf_{Y'}"'] & {[G(X)](Y'')} \arrow[d, "Gf_{Y''}"] \\
{[G(X')](Y)} \arrow[r, "G(X')g"] \arrow[d, "Gf'_Y"'] & {\tilde{G}(X', Y') := [G(X')](Y')} \arrow[r, "G(X')g'"] \arrow[d, "Gf'_{Y'}"'] & {[G(X')](Y'')} \arrow[d, "Gf'_{Y''}"] \\
{[G(X'')](Y)} \arrow[r, "G(X'')g"'] & {[G(X'')](Y')} \arrow[r, "G(X'')g'"'] & {\tilde{G}(X'', Y'') := [G(X'')](Y'')}
\end{tikzcd}$$

  Here, adjoining diagonal squares provide the composition $\tilde{G}(f', g') \circ \tilde{G}(f, g)$, while the edges of the largest square provides $\tilde{G}(f' \circ f, g' \circ g)$, the commutation shows the functoriality.

So, by routine check on the definition, we have $\Phi(\tilde{G}) = G$, showing surjectivity on the objects.

III. Naturality:

Let's skip this...

IV. Equivariance Version:

Assuming the first three statements, the remaining statement is to show: Φ as a functor commutes with G-action. Let $g \in G$, which given any $F \in \mathsf{Cat}(\mathcal{A} \times \mathcal{B}, \mathcal{C})$, we have $g \cdot F := g \circ F \circ g^{-1}$ being the conjugation action of G on this functor category (where the left g is the functor on $\mathcal{C}$, and the right g^{-1} is the diagonal g^{-1} functor on $\mathcal{A} \times \mathcal{B}$). Let $N := \Phi(F)$, and $N' := \Phi(g \cdot F)$. Then, notice $g \cdot F$ satisfies the following, for all $f : X \to X'$ in $\mathcal{A}$, and $g : Y \to Y'$ in $\mathcal{B}$:

$$N'_X(Y) = g \circ F \circ g^{-1}(X, Y), \quad (g \cdot N)_X(Y) = (g \circ N \circ g^{-1})_X(Y) = (g \circ N_{g^{-1}(X)})(Y)$$

$$= g \circ N_{g^{-1}(X)}(g^{-1}(Y)) = g \circ F(g^{-1}(X), g^{-1}(Y))) = g \circ F \circ g^{-1}(X, Y)$$

(Yeah, we also need to verify that they're identical on the morphisms of $\mathcal{B}$, but let's skip it here...)

Here, we do need to be careful on if g is a functor on the categories, or the functor on the functor categories (as the second $g \circ N$ is really g acting on the pointwise functor $N_{g^{-1}(X)}$ using conjugation).

This shows $N', g \cdot N$ are identical on the objects. Tp $\square$

God the proof is actually long as hell...

Definition 1.2. Given a space X, the chaotic topological category $\mathcal{E}X$ has object space X, morphism space $X \times X$, such that:

- Identity map $i : X \to X \times X$ by $x \mapsto (x, x)$ is continuous (also the same as diagonal morphism).
- The source map $s : X \times X \to X$ by $s(x, y) := x$ is continuous (also the same as first projection).
- The target map $t : X \times X \to X$ by $t(x, y) := y$ is continuous (also the same as second projection).
- Composition $\circ : (X \times X) \times_{s,t} (X \times X) \to X \times X$ by $\circ((x, y), (y, z)) := (x, z)$ is continuous.

Since there is only one morphism between any two objects, a morphism is equivalent to specify its source and target.

In particular, for any $x \in X$, the maps $\eta_x, \mu_x : X \to X \times X$ by $\eta_x(y) := (x, y)$, and $\mu_x(y) := (y, x)$ are continuous (as they're fixing one coordinate, while doing inclusion on the right). These also form natural isomorphisms $\eta : x \to X$, and $\mu : X \to x$ as follow:

$$\begin{tikzcd} x \arrow[r, "{(x,x)}"] \arrow[d, "{\eta_x(y)=(x,y)}"'] & x \arrow[d, "{\eta_x(z)=(x,z)}"] \\ y \arrow[r, "{(y,z)}"'] & z \end{tikzcd}, \qquad \begin{tikzcd} y \arrow[r, "{(y,z)}"] \arrow[d, "{\mu_x(y)=(y,x)}"'] & z \arrow[d, "{\mu_x(z)=(z,x)}"] \\ x \arrow[r, "{(x,x)}"'] & x \end{tikzcd}$$

This shows as a category, $\mathcal{E}X$ is equivalent to the terminal category $*$ (which only has one unique topological category structure).

Also, if $X = G$ is a topological group (including discrete ones!) then $\mathcal{E}G$ is isomorphic to the translation category of G (as all morphisms can be interpreted as a translation from one group element to the other).

Let Π be a topological group (intepreted as the category of single object $*$, with morphism space Π), then there is a group homomotphism $\Pi^{\mathrm{op}} \to \mathrm{Aut}_{\mathcal{T}\mathsf{Cat}}(\mathcal{E}\Pi)$, by right multiplication. So, all $g \in \Pi$ satisfies $g : \mathcal{E}\Pi \to \mathcal{E}\Pi$ by $\sigma \mapsto \sigma g$ (which is a homeomorphism on Π by definition of topological group), and morphism $(\sigma, \tau) \mapsto (\sigma g, \tau g)$ (again, this is the continuous diagonal action on $\Pi \times \Pi = \mathrm{Mor}(\mathcal{E}\Pi)$).

Proposition 1.3. *The orbit category $\mathcal{E}\Pi/\Pi$ is isomorphic to Π as topological category.*

Proof. It's clear on the object space, as $\mathrm{Ob}(\mathcal{E}\Pi) = \Pi$ as set, so $\mathrm{Ob}(\mathcal{E}\Pi)\Pi = *$.

The map on the morphism space is $(\sigma, \tau) \sim (\sigma', \tau')$ iff there exists $g \in \Pi$, such that $(\sigma, \tau) = (\sigma' g, \tau' g)$. Which, this is equivalent to say $\sigma\tau^{-1} = \sigma' g g^{-1} \tau'^{-1} = \sigma' \tau'^{-1}$. So, we see all $(\sigma, \tau) \sim (e, \tau\sigma^{-1})$, and all $g, g' \in \Pi$ satisfies $(e, g) \sim (e, g')$ iff $g^{-1} = g'^{-1}$, or $g = g'$. So, we see each (e, g) establishes a distinct equivalent class, hence providing a bijection between $(\Pi \times \Pi)/\Pi \cong \Pi$, by $(e, g) \mapsto g$.

Finally, to show this is an isomorphism as topological category, we see such identification of morphism is topologically isomorphic (because $\Pi \to \Pi \times \Pi \to \Pi$ by $g \mapsto (e, g) \mapsto g$ is a homeomorphism). Then, given (e, g) and (e, g') in the quotient, since the second one is equivalent to $(g, g'g)$, then we see the composition is $\circ((e, g), (g, g'g)) = (e, g'g)$, which has quotient $g'g$, which is equivalent to the composition of the image of (e, g') and (e, g), showing it's preserving the composition map. Hence, it's an isomorphism of categories. $\square$

Theorem 1.4. *Given a G-space X, a topological group Π (regarded as a trivial G-space, by $G \to e \to \mathrm{Aut}(\Pi)$), the quotient functor $p : \mathcal{E}\Pi \to \Pi$ induces an isomorphism if topological G-categories*

$$\mathsf{Cat}(\mathcal{E}X, \mathcal{E}\Pi)/\Pi \xrightarrow{\sim} \mathsf{Cat}_G(\mathcal{E}X, \Pi)$$

Proof. In terms of objects, we have $\mathsf{Cat}(\mathcal{E}X, \mathcal{E}\Pi) \cong \mathsf{Set}(X, \Pi)$

(To do). $\square$

2 Bicategories & 2-Categories

Here, we use 1 to denote

Definition 2.1. A *bicategory* is a tuple $(\mathcal{B}, 1, c, a, \ell, r)$, such that:

- The objects $\mathrm{Ob}(\mathcal{B})$ is a class, called 0-*cells* (also denoted $\mathcal{B}_0$).
- The *morphism/hom categories*: For each $X, Y \in \mathcal{B}$, there exists a category $\mathcal{B}(X, Y)$, with the objects called *1-cells* (denoted $\mathcal{B}_1$), and the morphisms are called *2-cells* (denoted $\mathcal{B}_2$).

I. Terminologies for 2-cells: The composition in each $\mathcal{B}(X, Y)$ is called *vertical compositions* (here, denote use $\odot$ to denote it), the identity morphisms in $\mathcal{B}(X, Y)$ are called *identity 2-cells*, while an isomorphism is called *invertible 2-cells*, with inverse called *vertical inverse*.

II. Horizontal Composition: For each 0-cells $X, Y, Z \in \mathcal{B}_0$, here exists a *horizontal composition functor*

$$c_{XYZ} : \mathcal{B}(Y, Z) \times \mathcal{B}(X, Y) \to \mathcal{B}(X, Z)$$

Where given 1-cells $f \in \mathcal{B}(X, Y)$, $g \in \mathcal{B}(Y, Z)$, and 2-cells $\alpha \in \mathcal{B}(X, Y)$, $\beta \in \mathcal{B}(Y, Z)$, we denote

$$c_{XYZ}(g, f) = g \circ f, \quad c_{XYZ}(\beta, \alpha) = \beta * \alpha$$

III. Associator: Given $W, X, Y, Z \in \mathcal{B}_0$, there exists *associator isomorphism*

$$a_{WXYZ} : c_{WXZ} \circ (c_{XYZ} \times \mathrm{Id}_{\mathcal{B}(W,X)}) \xrightarrow{\sim} c_{WYZ} \circ (\mathrm{Id}_{\mathcal{B}(Y,Z)} \times c_{WYX})$$

between the functors $\mathcal{B}(Y, Z) \times \mathcal{B}(X, Y) \times \mathcal{B}(W, X) \to \mathcal{B}(W, Z)$ (this is saying, composition of 1-cells are only up to isomorphism, and may be different from the hom class of $\mathcal{B}_0$).